ثانوية المدينة التعليمية
Education City High School

2.5

Exponential Function Context and Data Modeling

Student Notes and Guided Practice

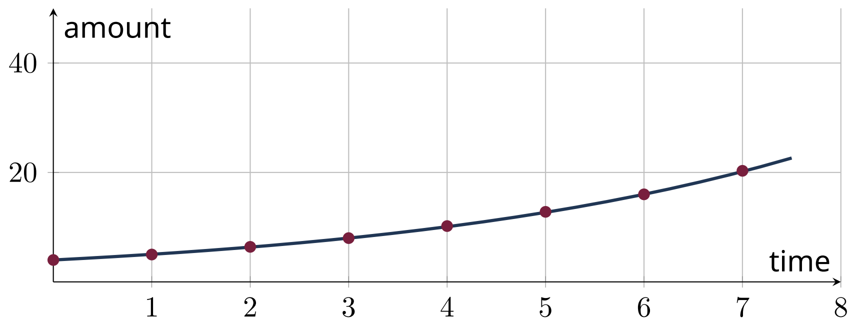
AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Construct exponential models from percent rates, two points, and regression.
- Interpret model parameters and equivalent growth factors.
- Use models to predict and solve inverse questions.

Essential Question

How can proportional change be encoded and interpreted in an exponential model?



Concept Framework

Key Idea

Initial value is the output at the chosen time origin.

Key Idea

A rate of r per period gives factor $1 + r$ or $1 - r$.

Key Idea

Two-point data determine the factor through a root when inputs differ by more than 1.

Key Idea

Predictions require appropriate units and a stated domain.

Formula Map

Formula and Structure

$$A(t) = A_0(1 + r)^t$$

Formula and Structure

$$A(t) = A_0 e^{kt}$$

Formula and Structure

$$k = \frac{\ln(A(t)/A_0)}{t}$$

Worked Examples

Worked Example 1

Problem. A culture starts at 500 cells and grows 18% per hour. Write a model.
Solution. $A(t) = 500(1.18)^t$.

Worked Example 2

Problem. A value falls from 80 to 50 in 6 years. Find the annual factor.
Solution. $b = (50/80)^{1/6} = (5/8)^{1/6}$.

Worked Example 3

Problem. Interpret $A(t) = 240e^{-0.07t}$.
Solution. Initial amount is 240; continuous decay rate is 0.07 per time unit; the model decreases toward 0.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Write an exponential model: initial 900, growth 6% yearly.

2. Write an exponential model: initial 240, decay 18% monthly.

3. Write an exponential model: initial 70, doubles every 5 units.

4. Write an exponential model: initial 1200, half-life 12 years.

5. Write an exponential model: initial 15, continuous rate 0.08.

6. A population is 5000 at t=2 and 7200 at t=6. Construct a model centered at t=2.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is ______________________.
2. A restriction I must remember is ______________________.
3. One graphical feature I can justify algebraically is ______________________.

Video support: Flipped Math topic page | Original ECHS material informed by Pearson and Holt Precalculus concepts.